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RESEARCH ARTICLE

Botany

The role of intraspecific variation in bumblebee body size and behavior on buzz pollination of a tropical legume species

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Abstract

Premise: The outcomes of generalized mutualisms rely on partner trait variation. In pollination mutualisms, although less explored, intraspecific variation in pollinator traits can be pivotal for successful pollination. We investigated the role of intraspecific body size and behavioral trait variations of bumblebee *Bombus morio* on the pollination of a buzz-pollinated legume species, *Chamaecrista latistipula*.

Methods: To explore the impact of body size and behavior of *B. morio* on the pollination of *C. latistipula*, we observed visits to virgin flowers and quantified the pollen removal and deposition (pollination performance) and fruit and seed production (reproductive fitness). By analyzing video and sound recordings, we measured *B. morio* body size and behavior on each flower, including bee vibration descriptors.

Results: We observed intraspecific behavioral differences among *B. morio* bumblebees associated with different body sizes. Larger bumblebees had half the handling time and vibrational pulses, less angular displacement within flowers, and larger relative peak amplitudes during vibrations than smaller bumblebees did. Regardless of their large variation in size and behavior, bumblebees were equally effective in removing pollen and pollinating flowers. The high female reproductive fitness was independent of bumblebee body size and behavior, likely due to the interaction between both. On the other hand, larger bumblebees visited flowers for shorter periods, probably promoting higher male reproductive fitness.

Conclusions: This study is the first to highlight the large intraspecific variation of bumblebee body size and behavior in buzz-pollinated flowers in the field. Together, body size and behavior effects unexpectedly cancel each other, generating a high buzz pollination efficiency.

KEYWORDS

Bombus morio, floral vibrations, handling time, mutualism, pollen foraging, pollination performance, poricidal anthers

The outcome of mutualistic interactions varies in space and time associated with trait variations in the partner species (Bronstein, 1994, 2015). Behavioral, morphological, and physiological differences between mutualistic partner species directly impact the balance and outcomes of these interactions (Stanton, 2003). Trait variation in mutualistic partners may partially explain the conditionality of these systems (e.g., Leal et al., 2023). For example, larger ant species disperse Euphorbiaceae seeds over greater distances than smaller ant species in generalized ant–plant mutualisms (Leal et al., 2014). Variations in ant body size, in this case, determine seed dispersal success, although the dispersal also depends on the amount of pulp around the seed. In another mutualistic system, *Allium vineale* plants increased carbon flow to roots, preferably hosted by fungal species that release relatively more phosphorus (Ji and Bever, 2016). However, plants decrease the carbon supply to the roots associated with the fungus in soils with high phosphorus availability (Ji and Bever, 2016). In both cases and others, the reciprocal exploitation of resources and services between species engaged in mutualisms can change due to the variation of partner traits and consequently define the outcome of these interactions for both interacting species (Heath and Stinchcombe, 2014; Dunkley et al., 2020, Leal et al., 2023).

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In pollination mutualisms, flowering plants rely on animal pollinators for sexual reproduction (Buchmann, 1983; Sakai, 2002; Switzer et al., 2016). Among these pollinators, bees provide pollination services globally (Michener, 2007; Rech et al., 2014). Certain bee species exhibit unique behaviors within specialized floral systems, such as floral vibrations on poricidal flowers (Delgado et al., 2023). In more than 20,000 species of plants with poricidal anthers (Vogel, 1978), pollen grains are enclosed within the anthers, being released only when specific bee species vibrate flowers during buzz pollination (Buchmann, 1983; Michener, 2007; Mesquita-Neto et al., 2018; De Luca et al., 2019). The design of poricidal anthers restricts pollen release (Buchmann, 1985; Harder and Thomson, 1989), increasing plant dependency on female vibrating bees for reproduction (De Luca et al., 2013; Vallejo-Marín, 2019). Hence, pollination efficiency can be mainly influenced by body size variation combined with behavioral differences during flower visits (Zych et al., 2013). Pollinator efficiency is expected to decline with decreasing body size (Armbruster et al., 1989). Smaller bees might not touch the stigma of relatively larger flowers while foraging for floral resources. In contrast, larger bees are more likely to touch the stigma because their size exceeds the spatial separation between the flower's sexual organs (Solís-Montero et al., 2015). In these cases, the morphological fit of pollinators and flowers is the first bound to regulate which floral visitors have the potential to pollinate plants (Armbruster and Muchhala, 2009). However, even touching the stigma, certain vibrating bees could encounter challenges in transferring pollen, particularly if their bodies carry a minimal pollen load in the regions that come into contact with the stigma. These factors together can determine the thresholds to distinguish pollinators from antagonists in buzz-pollinated plants (Solís-Montero et al., 2015, Mesquita-Neto et al., 2021).

In buzz-pollinated flowers, bee trait variation (e.g., body size and behavior) and its impact on pollination have been rarely explored (Table 1). Although both interspecific (e.g., Mesquita-Neto et al., 2021) and intraspecific (e.g., Solís-Montero et al., 2015) variation in bee body size can impact pollination, the interspecific bee body size variation is expected to be much higher than intraspecific ones. However, information scarcity precludes a thorough test of these expectations (Table 1). It inadvertently reduces the appreciation of the potential relevance of intraspecific variation in bee body size (e.g., Jauker et al., 2016). In this context, we present a pioneering study on the influence of intraspecific body size variation on bumblebee behavior during buzz pollination in the field and the implications for plant reproduction. Exploring the impact of size variability within pollinator species in the field is important because size variation is expected to be higher than in controlled laboratory experiments (Table 1). Intraspecific variation in bee body size could be associated with the complex behavior of vibrating female bees on poricidal flowers, impacting buzz pollination. In buzz pollination, handling time can be divided into the period when bees vibrate the poricidal anthers and extract pollen grains ("buzzing time") and the period when bees clean themselves and transfer pollen grains to their scopa or corbicula ("grooming time"; Lichtenberg et al., 2018). Vibrations transmitted from smaller bees have longer vibrational pulses ("buzzes") than larger bees with shorter vibrational pulses (De Luca et al., 2014). However, it is still unclear whether the duration of the grooming behavior is related to body size variation (Russell et al., 2021). Also, vibration behavior can be characterized by the fundamental frequency, amplitude, and duration (Cocroft et al., 2006; Sueur, 2018). Although negatively related to body size (De Luca et al., 2019), the fundamental frequency has little effect on pollen release, pollen collection and pollination compared to vibrational amplitude and duration (different from the handling time, the duration is the time expended in one buzz during the bee visit) (De Luca et al., 2013). Bees that vibrate flowers for longer periods (e.g., more pulses or buzzes) and with higher relative peak amplitudes remove more pollen grains from poricidal anthers (De Luca et al., 2013) and could transfer more pollen to subsequent flowers. Therefore, larger bees can spend less time per flower visit (Tavares et al., 2018) applying a higher amplitude vibration on poricidal anthers and, depending on the pollen fates, can increase the pollen transfer across plants than smaller bees (Willmer and Finlayson, 2014). All these aspects of bee behavior could create a gradient of pollination efficiency varying within and between bee species, which could be explained ultimately by bee body size variation.

Bumblebees (bees of the genus *Bombus*) play a central role in the pollination of numerous plant species, including many species with flowers with poricidal anthers (Morandin et al., 2001; Harter et al., 2002; Lye et al., 2011; Mesquita-Neto et al., 2018; Rosi-Denadai et al., 2020). Bumblebee worker body size can vary 10-fold within a colony and even more across colonies (Goulson et al., 2002; Austin and Dunlap, 2019), but the role of this huge body size variation on pollination has been poorly explored in the literature, especially under field conditions. Also, larger bumblebees are more likely to pollinate flowers (as explained above) because they can carry larger pollen loads on ungroomed body parts, increasing the chances of pollination during flower visits under natural conditions (Goulson et al., 2002; Willmer and Finlayson, 2014). In this study, we investigated how intraspecific body size variation in bumblebee workers, *Bombus morio*, commonly called *mamangavas* in Brazil, relates to their vibration behavior during foraging on flowers of a legume species. Also, we investigated whether body size and behavioral variation of *B. morio* determine the amount of pollen removed and deposited (pollination performance) and fruit and seed set (plant reproductive fitness). We chose the legume *Chamaecrista latistipula* (Benth.) Afr. Fern. & E. Nunes (Fabaceae) (Rando et al., 2020) as the focal plant species, in which *Bombus morio* bees are responsible for more than 60% of the flower visitations (Delgado, 2019). More specifically, we addressed three main questions: (1) What is the relationship between bumblebee body size and behavior during floral visits? (2) Does body size and bumblebee foraging behavior determine the amount of pollen removed from anthers and transferred to the stigma? (3) Do body size and bumblebee behavior

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TABLE 1 Summary of studies investigating the effect of bee body size on pollination and/or bee behavior on flowers, including study designs exploring the impact of bee body size within (intraspecific variation) and between species (interspecific variation). For the complete list of reviewed studies, see Appendix S16.

		Number of studies (%)			Percentage of studies		
Flower type	**Type of bee body size variation studied**	**Field**	**Lab**	**Bee body size metric measured[a]**	**Categorizing bee body size measurements**	**Linking bee body size with bee behavior on flowers[b]**	**On effect of bee body size on pollination success (positive effect)[c]**
Poricidal flowers (13 studies)	Intraspecific	**0 (0)**	5 (17.2)[d]	Head width, body mass, thorax width and length, abdomen width and length, and/or intertegular span	20.0	80.0	80.0 (50.0)
	Interspecific	8 (27.6)	0 (0)	Body length, wing length, mass, thorax width, and/or intertegular distance	62.5	87.5	87.5 (57.1)
Other flower types (16 studies)	Intraspecific	1 (3.4)	1 (3.4)	Bee height and/or intertegular span	50.0	50.0	100 (50.0)
	Interspecific	15 (51.7)	0 (0)	Body length, body width, thorax width, thorax length, thorax height, and/or intertegular distance	35.7	71.4	100 (57.1)

Notes: We searched for papers in the Web of Science database (http://www.webofknowledge.com/) using all available years up to December 2022. Our search used the following key terms: (bee* OR bumblebee*) AND (body size* OR "body length*" OR "bee size*" OR "bumblebee size*" OR "body-size*" OR "bee-size*" OR "pollinator size*" OR "intertegular distance") AND ("bee behavio*" OR "behavio* trait*: OR behavio*) AND (pollination* OR flower*) AND (success* OR efficienc* OR effective* OR "pollen transfer" OR "pollen deposition" OR "seed set" OR "fruit set" OR "plant reproduction" OR "plant sexual reproduction"). The search resulted in 83 papers. We read the abstracts and included only papers in which the effect of bee body size was evaluated on bee behavior on flowers and/or pollination success. We considered bee body size metrics described qualitatively (in categories) and quantitatively.

[a]Studies did not use the same metrics to describe the body size of bees, making comparisons across studies inconclusive.

[b]For behavior on flowers, most studies related body size variation only to flower handling time. However, studies that explored the effect of intraspecific body size variation in flowers with poricidal anthers also evaluated specific descriptors of bee vibration, such as fundamental frequency, amplitude, and duration of vibrational pulse.

[c]The positive effects of bee body size on descriptors of pollination success were predominant, regardless of whether the study investigated intra or interspecific variation in bee body size. Negative effects were extremely rare, while neutral effects were also common.

[d]Among the five studies that evaluated the role of intraspecific variation of bee body size on the pollination success of plants with poricidal anthers, all were experiments performed in a laboratory using a few bee colonies. On average, these studies used 14 colonies.

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determine fruit and seed formation? Because there are few studies investigating how intraspecific variation of the body size and behavior of wild bees affects pollination (Table 1) and none have been field studies, we used the results from studies on both intraspecific and interspecific body size and behavior variation and its effects on pollination to develop our predictions. In comparison to smaller *B. morio* bumblebees, we predicted that larger bumblebees would spend less time in flowers and consequently less time vibrating poricidal anthers and moving around within the flower. In addition, we expected that vibrations from larger bumblebees would generate higher relative amplitude and lower fundamental frequency. Finally, we expect that larger bumblebees would remove more pollen from poricidal anthers and transfer more to the stigma compared to smaller bees. Consequently, in this case, we anticipated that the resultant fruits would contain more viable seeds.

MATERIALS AND METHODS

Species and study site

We chose a leguminous plant species belonging to the genus *Chamaecrista* and a bumblebee species belonging to the genus *Bombus*, both widely distributed throughout Brazil (Françoso, 2015; Souza and Bortoluzzi, 2015). *Chamaecrista latistipula* (Figure 1A, B), previously known as *Chamaecrista desvauxii* var. *latistipula* (Rando et al., 2020), is an endemic shrub widely distributed from the northern to the southern regions of Brazil (Irwin and Barneby, 1982). The flower diameter is around 2 cm with five sepals and five petals with one larger and more curved called the cucullus (Nogueira et al., 2018; Delgado, 2019). The flower has an heterantherous androecium with three larger and seven short fertile stamens with poricidal anthers (Gottsberger and Silberbauer-Gottsberger, 1988; Nogueira et al., 2018) and a unicarpelar gynoecium enveloping the ovules. *Chamaecrista latistipula* is self-compatible and produces flowers nearly all year, with a flowering peak typically between January and March (Valadão-Mendes, 2018). This pattern is also observed in the population found at the Universidade Federal do ABC (UFABC; https://www.ufabc.edu.br/en/), where the study was conducted.

The genus *Bombus* of the family Apidae has approximately 250 medium-sized bumblebee species distributed worldwide (Michener, 2007). In Brazil, six species, including our focal bumblebee *Bombus morio* (Figure 1C, D, E, F), occur from northwestern South America to southeastern Brazil, Uruguay, and northern Argentina (Benavides, 2008; Françoso, 2015). *Bombus morio* is black, hairy, and its body length ranges from 9 to 22 mm (Michener, 2007; Françoso, 2015). The females are common foragers of pollen flowers due to their ability to vibrate poricidal anthers (Buchmann, 1983). This species is a common visitor to *Chamaecrista* flowers throughout the Cerrado Domain (Gottsberger and Silberbauer-Gottsberger, 2006; Mesquita-Neto et al., 2018), reaching an average of 10

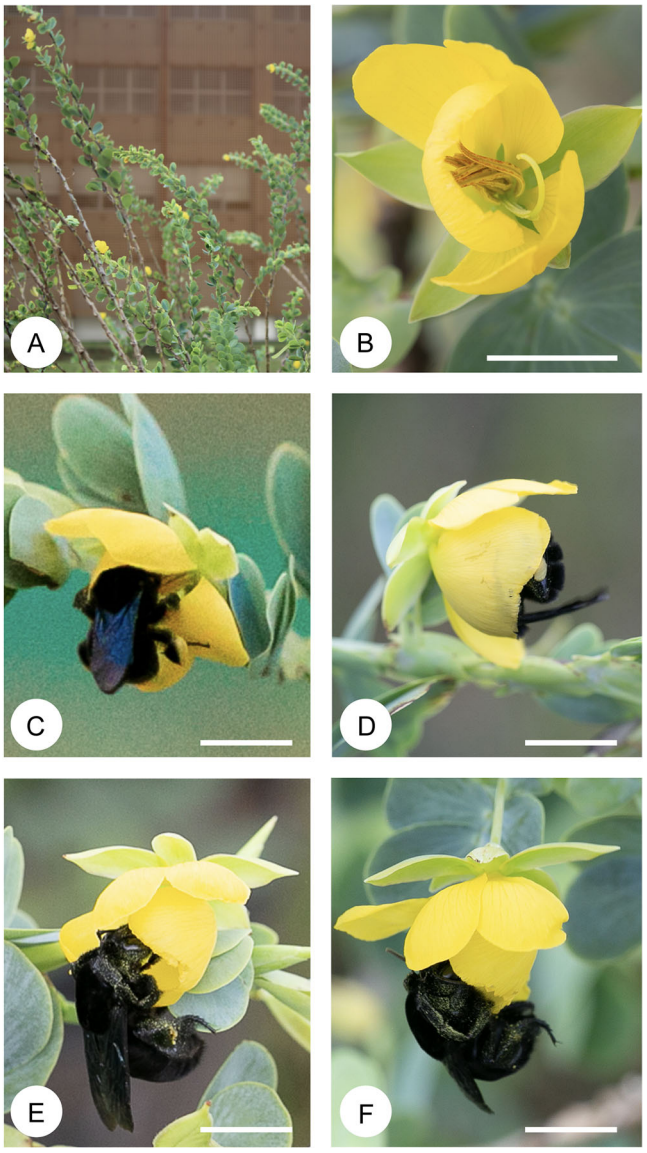


FIGURE 1 *Chamaecrista latistipula* flower and visits by *Bombus morio* to flowers in the study population. (A) Floral display; (B) flower details; (C, D) *B. morio* individual with a smaller body size, and (E, F) an individual with a larger body size. Scale bars: (B) 2.0 cm and (C–F) 1.0 cm.

visits per hour to flowers of *C. latistipula* (Nogueira et al., 2018). Indeed, each *C. latistipula* flower in our experimental population is typically visited by three bees from different species within the first 30 min after flower opening, with *B. morio* being the most frequent visitor. In these cases, visits result in an accumulation on average of 59 ± 20 pollen grains on the stigma (Delgado, 2019). Although bumblebees usually wrap its body, holding the anthers with their mandibles (e.g., Morgan et al., 2016), our preliminary observations on *C. latistipula* flowers indicated that *B. morio* do not hold the anthers with their mandibles. Instead, with its head close to the base of the anthers, this bumblebee uses its legs and bristles to hold onto the floral structure, especially petals, and tap the base of the poricidal anthers, transfering the vibrations produced by contracting its thoracic muscles (A. Nogueira,

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personal observations; but see Switzer et al., 2016). In other words, the body of the bee is inverted relative to the apical anther pores (Nogueira et al., 2018).

The study area on the Universidade Federal do ABC (UFABC) campus in the municipality of São Bernardo do Campo, in the São Paulo state, Brazil, covers 2500 m^2 and contains 30 *C. latistipula* shrubs transplanted in December 2018 from a shoreline population near the Cantareira Reservoir in Franco da Rocha (Appendix S1). The shrubs were transplanted into 18-L pots at the peak of the rainy season to prevent posttransplant mortality. Most of the original soil associated with shrub roots was preserved, and a mixed substrate of 70% fine-washed sand and 30% topsoil (v/v) was used to complete the pots.

Sampling design: bumblebee size and behavior during flower visitations

We observed bumblebee visitation to virgin *C. latistipula* flowers to investigate the relationship between body size and behavior of the bees and their impact on the pollination performance and plant reproductive fitness (see Opedal et al. [2023] for the definitions and components of pollination performance and reproductive fitness). We selected 5–10 flower buds from each of the 30 individuals of *C. latistipula* and covered them with small tulle bags to prevent visits by bees and other insects. Working with one flower at a time as anthesis began, we removed the tulle bag at 06:00 hours and allowed only one *B. morio* individual to visit the focal flower. After the visitation, the flower was bagged again to prevent further visits. If the first visitor to the flower was not a *B. morio* bee, then the flower was discarded from our sample. Field sampling was done on nonconsecutive days from 25 January 2021 to 4 March 2021.

We filmed and recorded the sound of each bumblebee visiting a flower. We filmed the visits with a Canon (Ota, Japan) T5i 18–55 lens camera placed on a tripod and positioned to the level of the flower (the height at which the camera is held relative to the ground). Alongside each flower, a scale was placed at a standardized distance and position relative to the camera. The scale was fixed on the plant itself and parallel to the flower, 20 cm from the camera lens, in frontal view. From the video recordings, we captured images of the bumblebees in the flowers and estimated thoracic and abdominal widths and intertegular distance (i.e., the distance across a bumblebee's thorax between the bases of the wings) using ImageJ 1.8.0 software. We described the behavior of bumblebees by analyzing their movements within the flowers. For this, we quantified the angular displacement of *B. morio* individuals as they moved around the anthers after each vibrational pulse. These movements resulted in circular displacements around the central poricidal anthers. However, because bumblebees moved clockwise and counterclockwise during the same visit, we developed two other angular movement variables to describe their complex foraging behavior. In summary, we measured three bumblebee displacements (in degrees): (1) the maximum angular displacement, (2) the summed angular displacement, and (3) the mean angular displacement between vibrational pulses. The maximum angular displacement refers to the furthest angular position in which the bumblebee performed the vibrational pulses around the central anthers to the position of the first vibrational pulse. The summed angular displacement is the sum of a bumblebee's circular movements, including clockwise and counterclockwise displacement, between vibrational pulses around the central anthers. The mean angular displacement between vibrational pulses is the summed angular displacement divided by the number of vibrational pulses (buzzes) subtracted by one (the position of the first vibrational pulse in the flower is set as position zero) (Appendix S2). In addition, we measured other variables describing the bumblebee behavior during flower visits, i.e., the number of vibrational pulses, handling time (the total time the bumblebee spent on the flower in seconds), buzzing time (the time spent vibrating the poricidal anthers in seconds), and grooming time (the time bumblebees clean the pollen grains for their bodies in seconds). The handling time necessarily includes the buzzing and grooming times. For more details on these behavioral measurements, see Appendix S3.

The audio recordings of each bee visit were made with a CSR Yoga HT-81 unidirectional microphone and a Sony ICD-PX240 recorder (Sony, Tokyo, Japan) on a tripod next to the camera, positioned at the level of the flower at a standardized distance of 15 cm from the flower. Also, the tip of the unidirectional microphone was placed at a standardized orientation relative to the flower in all cases. Using the software Audacity 2.3.2 (open-source software, https://www.audacityteam.org/), we analyzed the acoustic recordings of each bumblebee visit to the focal flowers. For the acoustic analysis, we initially standardized the spectrogram for size 8192, "Hamming" type, at a maximum frequency of 3000 Hz with grayscale, a high-pass filter with a roll-off of 12 dB per octave, and a cut-off frequency of 100 Hz for noise reduction. We quantified the sound of bumblebees during flower visits using two variables that describe the vibrations: the fundamental frequency (in Hz) and the peak relative amplitude (dB). To obtain single estimates of the two variables for each bumblebee visit to a flower, we averaged the fundamental frequencies and peak relative amplitudes measured from all of the individual vibrational pulses emitted by the bumblebee during a flower visit. According to De Luca et al. (2020), the acoustic amplitude does not represent the vibrational amplitude accurately. It can only be used as a comparative measure between bees when identical procedures are followed in each recording, such as gain level settings, fixed position, and distance between the microphone and the observed flower. We followed these procedures for a faithful extraction of this information. Hereafter, when we mention fundamental frequency and peak relative amplitude, we refer to the average of these metrics among vibrational pulses within a single flower visit. Using the videos of the flower visits, we broke down the

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duration (in seconds) of bumblebee behaviors into handling, buzzing, and grooming time, as described above.

We also applied the morphological size-matching index proposed by Solís-Montero and Vallejo-Marín (2017) to evaluate whether bumblebee body size relative to the spatial separation between floral sexual organs influenced the probability that bumblebees contacted both anthers and stigmas during flower visitation. These data were obtained using a caliper to measure the herkogamy of each flower, that is, the distance between the stigma and the pore of the anther closest to the gynecium (spatial separation of floral sexual organs within a flower; Barrett, 2002). The size-matching index is the difference between the herkogamy measurement of the flower and the width of the bumblebee's abdomen (Solís-Montero and Vallejo-Marín, 2017). When the difference between the herkogamy of the flower and the width of the bumblebee abdomen is zero, the bumblebee abdomen fits exactly in the space between the stigma and the anther pore. When the size-matching index is positive, the herkogamy is larger than the bumblebee's abdominal width, so the bumblebee cannot simultaneously touch both floral sexual organs. When the size-matching index is negative, the herkogamy is smaller than the bumblebee's abdominal width, so the bumblebee touches both floral sexual organs simultaneously.

Pollen removal, transfer, and fruit and seed formation after *Bombus morio* visits

To evaluate pollen removal (male pollination performance component) from the poricidal anthers by bumblebees of contrasting sizes and behaviors, we quantified the pollen remaining on the anthers after each visit. For this evaluation, we assumed that the quantity of pollen was similar among the flowers before the bumblebee visits. We quantified the pollen by carefully removing the androecium of the visited flowers and fixing them in 50% FAA (5 mL formalin + 5 mL acetic acid + 90 mL 50% ethanol) for 24 h. We placed each androecium in a 2-mL microcentrifuge tube and added 600 μL of distilled water with Tween 20 (solution prepared from 250 mL of distilled water with a drop of Tween 20). We added a 0.90-g microsphere and mixed the contents of the tube using a vortex mixer (BiomiXer model VTX-2500, Benchmark Scientific, Edison, NJ, USA) at maximum speed for 180 s. While keeping the tube in the mixer to maintain the pollen grains in suspension, we removed 10 μL from the solution with a micropipette and placed it in a Neubauer chamber to count the pollen grains with a light microscope at 40× magnification. This procedure for extracting and counting pollen grains was ideal given the size of the anthers, anther pores, and pollen grains of *C. latistipula*. The protocols for pollen collection, fixation, dilution, and estimation, and checking for potential pollen counting errors, specifically underestimations, are detailed in Appendix S4. To estimate the total number of pollen grains in *C. latistipula* anthers before bumblebee visits, we used the same techniques to count pollen in virgin anthers of 11 floral buds from 11 different plants of *C. latistipula*. We quantified the number of pollen grains removed during each bumblebee visit by calculating the difference between the number of pollen grains in unvisited virgin flowers and the number of pollen grains counted in the flowers visited by bumblebees.

To evaluate pollen deposition (female pollination performance) by bumblebees with contrasting sizes and behaviors, we quantified the pollen grains on the stigma of each flower by detaching the distal portion of the style 24 h after bumblebee visitation. For that, we bagged and labeled each flower visited by *B. morio* before the time required to separate the stigma from the flower. The rest of the gynoecium remained attached to the plant, allowing fruit development. We tested this procedure during a previous pilot study. Removing the distal portion of the style 24 h after depositing the pollen on the stigma does not prevent fruit and seed formation because pollen tubes form and reach the ovules within the first 12 h after pollination (B. Campos Barbosa and G. V. Mota, personal observations). In addition, the counts of pollen grains on the stigma did not differ after 24 h. The distal portion of the style removed from the flowers was fixed in 50% FAA solution for 24 h, then transferred to 70% v/v ethanol. To quantify the pollen grains deposited on the stigma, we removed the section of style most distant from the stigma and transferred the sample to a microscope slide. A drop of carmine acetic acid and a cover slip were placed on the tip of the stigma sample (Appendix S5A). After 5 min, we observed the material with a light microscope at 40× magnification (Appendix S5B).

To evaluate fruit and seed production (female reproductive fitness, according to Opedal et al., 2023) after bumblebee visits, we inspected the plants and flowers once a week for fruit. Once fruit development began, we enclosed the immature fruit in a small tulle bag, and after its maturation (when it became dark and rattled), we collected the fruits. In the laboratory, we counted well-formed seeds and those that did not develop completely.

Statistical analyses

To investigate the relationship between the size and behavior of bumblebees, we initially applied a multivariate ordination technique to the nine quantitative variables describing their behavior on flowers. Because ordination analysis assumes that some of the variables are redundant, we applied the Bartlett sphericity test to assess the degree of correlation among the nine behavioral variables, i.e., handling time, buzzing time, grooming time, number of pulses, fundamental frequency, peak relative amplitude, and maximum, summed and mean angular displacement. The results showed significant covariance between variables ($\chi^2 = 587.06$, $P < 0.01$), indicating that we could use an ordination technique such as a principal component analysis (PCA). Before applying the PCA, we standardized all variables using the scale function in the base

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packages of R (R Core Team, 2021) to prevent the different units of the variables from interfering with the principal components. After applying the PCA, we verified which principal components and loadings were biologically interpretable using randomization tests proposed by Björklund (2019), using the randomizeMatrix function in the R package spicy (Kembel et al., 2010). Once the relevant PCA axes were determined, we tested the relationships between the axes and the variables describing bumblebee body size variation using simple linear regression models. For the regression analyses, we tested assumptions of normality and homogeneity of variance of the residuals using visual inspections of quantile–quantile (QQ) plots and residual plots. In cases where the assumptions were not met, we square-root transformed the values for the analyzed variables.

To test the role of bee body size and behavior (represented by PCA axes) on pollen removal, deposition, and fruit and seed formation, we constructed a set of eight generalized linear models (GLM). Because both PCA axes (PC1 and PC2) describing bee behavior were related to bee size (see Results), we did not use the PCA axes and bee size in the same GLM models. This procedure avoided the multicollinearity between predictor variables in the same model. The response variables used in these models were the proportion of pollen grains removed from the anthers by bees (proportion-type variable), the number of pollen grains deposited on the stigma (count-type variable), fruit formation (binary variable), and the proportion of viable seeds in the fruits (proportion-type variable). In the analyses using proportion-type variables, we used the binomial error distribution in which the response variables were treated as the number of failures (e.g., pollen grains remaining in the poricidal anthers) and successes (e.g., pollen grains removed by the bee) as a two-vector response variable, as suggested by Crawley (2007). Similarly, we used a GLM with a binomial probability distribution when analyzing the proportion of viable seeds per fruit. For pollen deposition (number of pollen grains on the stigma), we used a GLM associated with the negative binomial probability distribution due to the overdispersion of residuals compared to the application of the Poisson distribution. In this case, we applied the glm.nb function in the R package MASS (Venables and Ripley, 2002). To analyze fruit formation, we used a GLM associated with a binomial probability distribution. In all cases, the nature of the response variable and the degree of residual overdispersion determined the probability distribution applied for each statistical model.

We also applied the scale function to the variables before the analyses to standardize the estimation of the slope coefficients in each model. This procedure allowed direct comparison of the effect sizes of each factor, regardless of the unit of measure. Previously, we also tested whether the arrival time of the bumblebees at the flower impacted their behavior during visits. For all tests, bumblebee arrival time was irrelevant to the phenomena investigated here. All statistical analyses were done with RStudio Software (R Core Team, 2021).

RESULTS

Overall, 59 single visits by *Bombus morio* to 59 different virgin flowers were recorded, on average two flowers per plant. Body size measurements of *B. morio* visiting *C. latistipula* flowers were 8.9 ± 2.0 mm (mean ± SD) for thoracic width, 9.7 ± 2.3 mm for abdominal width, and 6.1 ± 1.7 mm for intertegular distance (i.e., the distance between the bases of the bumblebee's wings). Body size measurements were highly correlated with each other. Bumblebee visitors emitted, on average, 14.1 ± 10.8 (mean ± SD) vibrational pulses per visit. Mean ± SD bumblebee handling time was 37.2 ± 33.8 s, buzzing time was 15.1 ± 12.5 s (sum of individual pulses), and grooming time was 14.6 ± 16.2 s. In terms of bumblebee vibrational behaviors, mean (±SD) fundamental frequency was 198.7 ± 30.9 Hz and relative amplitude peak was −30.8 ± 5.7 dB. On average, vibrational pulse lasted 1.6 ± 1.5 s. The mean (±SD) maximum angular displacement was 92.2° ± 64.5°, and mean angular displacement between vibrational pulses was 25.1° ± 23.8°.

The bumblebees removed, on average ±SD, 883,774 ± 18,792 pollen grains from *C. latistipula* flowers, representing 98 ± 0.02% of the pollen in the anthers. The mean (±SD) number of pollen grains deposited on stigmas per visit was 39.8 ± 24.5%, and 88 ± 0.3% of the visits resulted in fruit formation. Each formed fruit had a mean ±SD of 11.0 ± 6.4 viable seeds, equivalent to about 59% of total ovules. The variability in the number of pollen grains on stigmas did not influence the subsequent formation of fruits or the number of viable seeds (Appendix S6). Of the 59 bumblebee visits to *C. latistipula* flowers recorded during this study, 58 showed negative size-matching index values, and only one showed a positive value. Negative index results indicated that the bumblebee's abdomen size could simultaneously touch both floral sexual organs (i.e., the distance between the anther and stigma was smaller than the abdominal width).

What is the relationship between bumblebee body size and their behavior during flower visitation?

The PCA of the variables related to bumblebee behavior (see summary of results in Appendices S7 and S8) indicated that the first two axes (PC1 and PC2) captured important biological information (randomization test: $P < 0.001$ for the first two axes; Appendix S9) and explained 54.2% and 20.7% of the variance, respectively, in behavioral metrics (Figure 2). Flower handling time, grooming time, buzzing time, the number of vibrational pulses per visit, maximum angular displacement, and summed angular displacement were the variables that contributed the largest loadings to PC1 (randomization test for each of the variables: $P < 0.001$; Appendix S10A). The relative amplitude peak and the mean angular displacement contributed the largest loadings to PC2

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(randomization test for each variable: $P < 0.001$; Appendix S10B). Fundamental frequency was the only bumblebee behavioral variable that did not significantly influence PC1 and PC2.

Bumblebee body size was negatively related to PC1 and PC2 (Table 2, Figure 3). PC1, representing six behavioral variables, had a stronger relationship with the thoracic width than the other body parts measured (first row of Table 2; Figure 3A). In contrast, PC2, representing two behavioral variables, had a stronger relationship with abdominal width (fifth row of Table 2; Figure 3D). Intertegular distance was the only body measure unrelated to the bumblebee behavioral variables (Appendix S11). Considering the loadings associated with each principal component and the relationships detected with the bee-size descriptors, larger bumblebees had shorter handling times, less buzzing and grooming, and emitted fewer vibrational pulses than smaller bumblebees. In addition, the rotational displacement of larger bumblebees around the flower anthers was less compared to smaller bumblebees, and larger bumblebees produced higher peak relative amplitude vibrations.

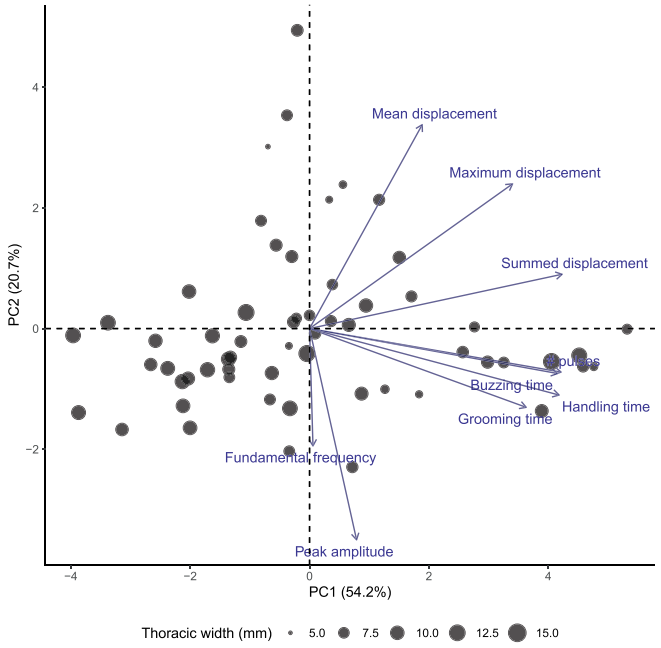


FIGURE 2 Principal component analysis (PCA) of the variables that comprise *Bombus morio* behavior during visits to *Chamaecrista latistipula* flowers. PC axes 1 and 2 explained 54.2% and 20.7% of the variance, respectively. The behavioral variables of *B. morio* included fundamental frequency, peak relative amplitude, grooming time, buzzing time, handling time, number of vibrational pulses, and mean, maximum, and summed angular displacement. Point sizes differ concerning the thoracic width of *B. morio* individuals (millimeters).

Are bumblebee body size and vibrational behavior related to pollen removal and transfer, and consequently, to fruit and seed formation?

Regardless of body size and behavior variation, *B. morio* removed 98 ± 0.02% of pollen grains from poricidal anthers and were very effective pollinators of *C. latistipula*. Thoracic width as a measure of bumblebee body size was not related to the pollen removal, pollen deposition, and fruit and viable seed formation (Table 3, Figure 4). Furthermore, the PC1 and PC2 axes, representing bumblebee behavioral traits, were also unrelated to those four variables describing the pollination performance and reproductive fitness (Table 3; Appendices S12–S15).

DISCUSSION

We investigated how *B. morio* bumblebee body size and behavioral variation impact the pollination in *C. latistipula* flowers. Our results showed a wide variation in the body size of this bumblebee species visiting flowers of *C. latistipula*. Negative values of the size-matching index indicated that virtually all bees could touch the flower stigma regardless of their size. However, using the index alone, it is challenging to verify pollination performance and reproductive fitness (Opedal et al., 2023). Interestingly, the behavior of these bees was complex and variable on

TABLE 2 Relationships between PC axes describing *Bombus morio* behavioral variables (response variables) and body size (in millimeters), represented by thoracic width, abdominal width, and intertegular distance measurements. The estimated coefficient is the standardized slope coefficient (b_{st} ± SE) describing the size effect of bee body size on behavioral axes. PC1 was most strongly explained by thoracic width (b_{st} = –0.65 ± 0.28). PC2 was most strongly explained by abdominal width (b_{st} = –0.52 ± 0.16). N = 59 flowers.

Response variables (Behavioral axes)	Predictor variables (Bumblebee size)	Estimated coefficient ± SE	*F*	*P*	Graph
PC1	Thorax width	–0.65 ± 0.28	5.5	0.02	Figure 3A
	Abdominal width	–0.55 ± 0.28	3.7	0.058	Figure 3C
	Intertegular distance	–0.48 ± 0.28	2.8	0.097	Appendix S11A
PC2	Thorax width	–0.39 ± 0.17	5.3	0.02	Figure 3B
	Abdominal width	–0.52 ± 0.16	10.1	0.002	Figure 3D
	Intertegular distance	–0.32 ± 0.17	3.4	0.069	Appendix S11B

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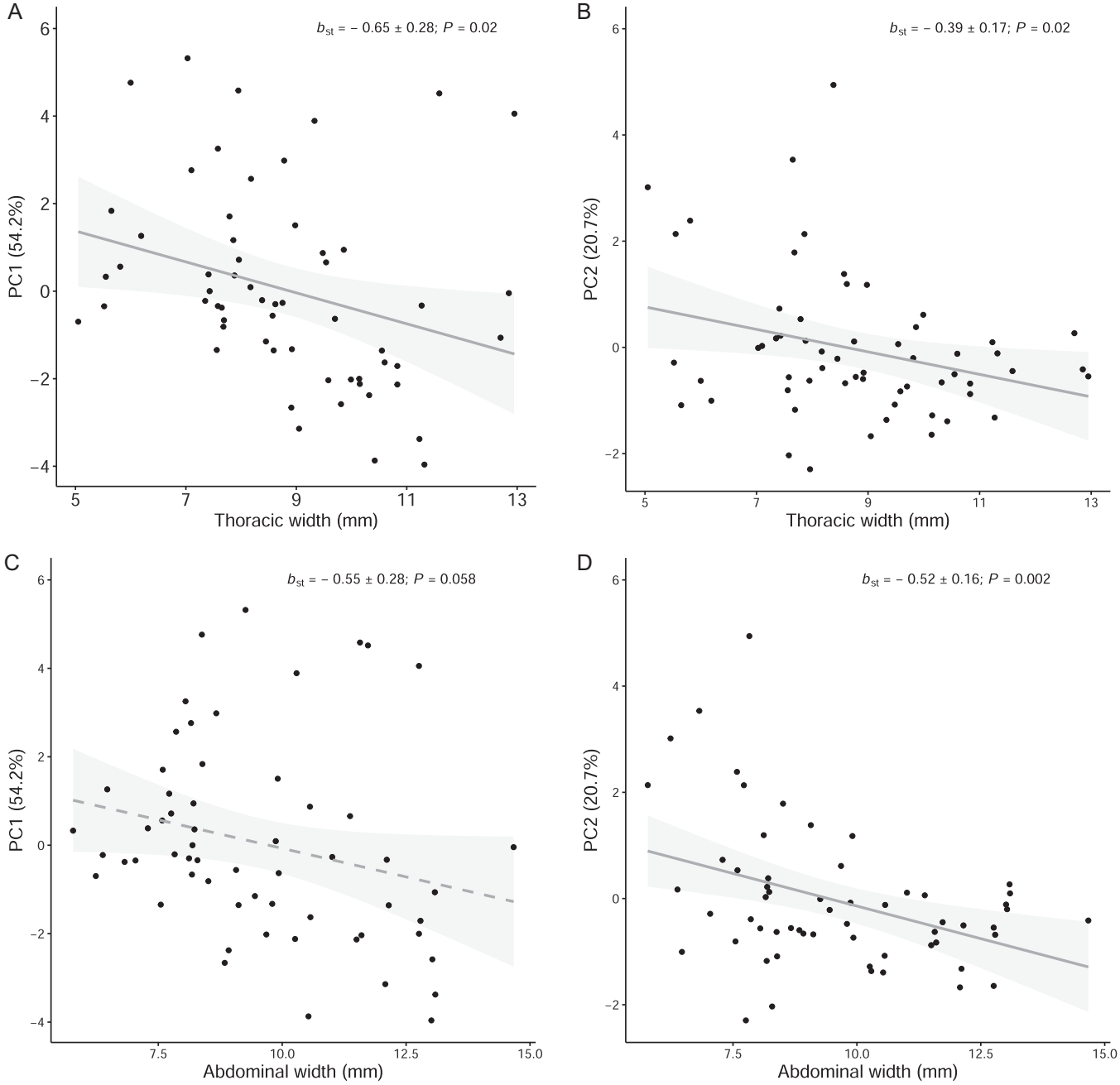


FIGURE 3 Relationships between *Bombus morio* body size and principal component axes PC1 and PC2 describing bumblebee behavior. (A) Relationship between thoracic width and PC1. (B) Relationship between thoracic width and PC2. (C) Absence of relationship between abdominal width and PC1. (D) Relationship between abdominal width and PC2. The dashed line represents the absence of a relationship between variables ($P > 0.05$), while the solid lines represent verified relationships between variables ($P < 0.05$). The gray shading represents the confidence interval around trend lines. The standardized slope coefficient (b_{st} ± SE) describes the size effect of bee body size variables on behavioral axes (PC1 and PC2).

flowers across individuals and was linked to variations in their body size, potentially influencing the pollination process. Indeed, in agreement with our expectations, we discovered that larger bumblebees (1) had shorter visit durations and spent less time handling the flowers, vibrating the poricidal anthers, and grooming themselves; (2) moved less in terms of angular displacement within a flower and between each vibrational pulse; and (3) vibrated with a higher peak relative amplitude compared to smaller bumblebees. Contrary to our predictions, all bumblebees were highly effective in pollinating flowers, including the smaller ones. Regardless of body size, bumblebees removed nearly all pollen grains from the poricidal anthers during single visits and deposited ample pollen onto the stigmas, triggering the production of fruits with numerous viable seeds. Our study demonstrated that bumblebees of different

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TABLE 3 Relationships between *Bombus morio* body size and behavioral variables associated with pollination performance (pollen deposition and pollen removal) and female reproductive fitness (fruit formation and proportion of viable seeds per fruit) and the removal of pollen grains from *Chamaecrista latistipula* anthers. Table rows without shading are results from models combining behavioral variables (PC1 and PC2). Shaded rows are results from models using body size (mm). The estimated coefficient is the standardized slope coefficient (b_{st} ± SE) describing the size effect of bumblebee traits on pollination performance or reproductive fitness.

Response variables (Pollination performance and reproductive fitness)	Predictor variables (body size and behavior)	Probability distribution	Estimated coefficient ± SE	*z* or *t*	*P*	Graph
Pollen deposition on stigma	PC1	Negative binomial	−0.02 ± 0.08	−0.27	0.79	Appendix S12A
	PC2		−0.06 ± 0.08	−0.72	0.47	Appendix S12B
Pollen deposition on stigma	T	Negative binomial	0.09 ± 0.08	1.22	0.22	Figure 4A
Fruit development	PC1	Binomial	−0.45 ± 0.41	−1.08	0.28	Appendix S13A
	PC2		−0.38 ± 0.36	−1.06	0.29	Appendix S13B
Fruit development	Thorax width	Binomial	0.59 ± 0.45	1.32	0.18	Figure 4B
Viable seeds	PC1	Binomial[a]	−0.07 ± 0.19	−0.37	0.71	Appendix S14A
	PC2		0.07 ± 0.20	0.35	0.72	Appendix S14B
Viable seeds	Thorax width	Binomial[a]	0.29 ± 0.19	1.53	0.13	Figure 4C
Pollen removal	PC1	Binomial[a]	−0.08 ± 0.10	−0.74	0.46	Appendix S15A
	PC2		−0.11 ± 0.09	−1.15	0.25	Appendix S15B
Pollen removal	Thorax width	Binomial[a]	0.04 ± 0.10	0.38	0.71	Figure 4D

[a]Proportion data were analyzed using logistic regression with binomial errors, as proposed by Crawley (2007), using two-column object for the response variable combining two vectors that contain the counts of success (e.g., viable seeds per fruit) and failure.

sizes had equal success in pollinating *C. latistipula* flowers. This phenomenon becomes evident through the interaction of bee body size and behavior with flowers, given their corresponding size match. In this scenario, variations in behavior linked to size differences could be explained by a compensatory mechanism that enables smaller bees to optimize pollen collection, thereby depositing an equivalent number of pollen grains onto the stigmas. Below, we discuss the relevance of these patterns in the reproduction of buzz-pollinated plants.

Smaller bumblebees move around more and emit less-intense vibrational pulses within flowers

Body size variation of *B. morio* was related to behavioral differences during flower visits. Smaller bumblebees spent an average of 40 s more manipulating the flowers (i.e., longer handling times) than larger bumblebees. This pattern resembled findings from other studies comparing bee species with varying body sizes (Vivarelli et al., 2011; Quinalha et al., 2017). For example, in *Jacaranda caroba* flowers, smaller bee species spent approximately 30 s longer in flowers than larger bee species (Quinalha et al., 2017). In our study, larger bumblebees emitted shorter-duration vibrational pulses in *C. latistipula* anthers, whereas the opposite was found for smaller sized bee species. Another buzz pollination study involving two *Bombus* species found a negative relationship between thoracic width and the duration of vibrational pulses in flowers. In this case, the *Bombus* species with a thorax 1 mm wider had vibrational pulses that were 0.33 s faster than those of the smaller species (De Luca et al., 2014). This pattern was similar to the intraspecific variation we observed in *B. morio* during our study. Although the relationship between grooming time and body size has not been widely investigated (e.g., Russell et al., 2021), our study showed that as *B. morio* body size increased, grooming time during floral visits decreased. Smaller bumblebees spent around 20 s more on grooming than the larger bumblebees. In addition, smaller bumblebees moved more within the flowers, with greater angular displacement around the poricidal anthers between vibrational pulses, in comparison to larger bumblebees. This difference likely occurred because smaller bumblebees were likely not to be able to vibrate all anthers together during a single pulse, whereas larger bumblebees could effectively embrace most of the androecium. A similar pattern was observed by Mesquita-Neto et al., (2021) across bee species of different sizes visiting *Chamaecrista ramosa* flowers. Larger bees, such as *Eulaema nigrita*, grasped all the anthers, simultaneously vibrating and nearly immobilizing them within the flower. In contrast, smaller bees, like *Melipona quinquefasciata*, typically vibrated the anthers one at a time as they moved among different positions (Mesquita-Neto et al., 2021). For smaller bees, the longer handling times, combined with longer duration vibrational pulses and greater movements within the flowers,

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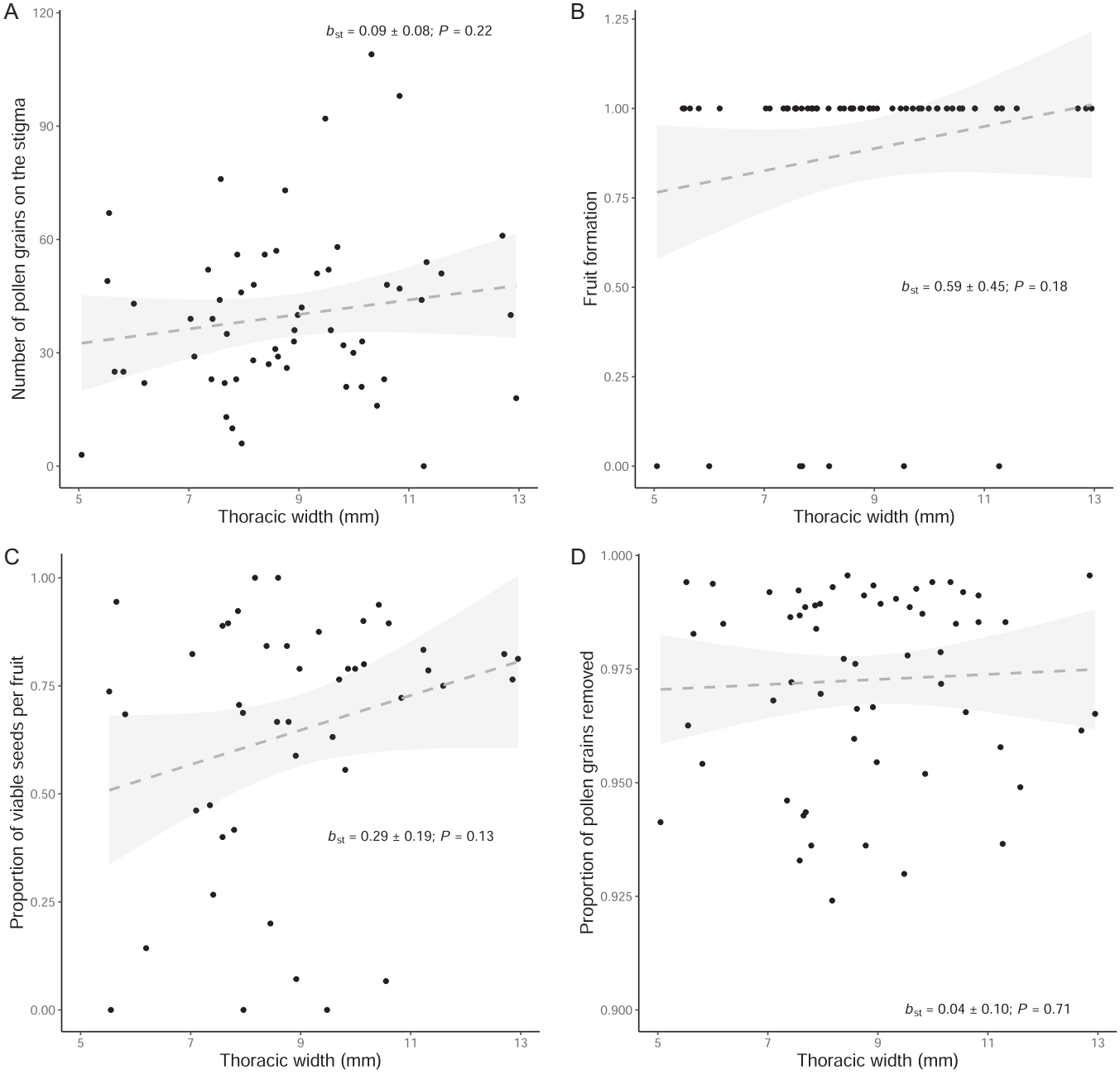


FIGURE 4 Relationships between the thoracic width (mm) of *Bombus morio* and pollination performance and female reproductive fitness of *Chamaecrista latistipula*. (A) Absence of relationship between thoracic width and the pollen deposition on the stigma. (B) Absence of relationship between thoracic width and fruit formation. (C) Absence of relationship between thoracic width and the proportion of viable seeds in the fruits. (D) Absence of relationship between thoracic width and the proportion of pollen grains removed from anthers. The dashed lines represent the absence of a relationship between variables ($P > 0.05$); the gray shading shows the confidence interval around the line. The standardized slope coefficient (b_{st} ± SE) describes the size effect of bumblebee traits on pollination performance or reproductive fitness.

potentially compensated for their smaller size, allowing them to achieve levels of pollen release and deposition similar to larger bee species (De Luca et al., 2013; Rosi-Denadai et al., 2020).

Supporting the findings of De Luca et al. (2013), our results showed that vibration amplitude, denoting the strength with which bees contract their thoracic muscles during vibration (King, 1993), was greater in larger *B. morio* bumblebees during flower visits. In a buzz-pollination system involving *Bombus impatiens* and *Solanum* flowers, larger *B. impatiens* individuals had vibrations with peak relative amplitudes 0.25 times higher, on average, than those of smaller *B. impatiens* individuals (De Luca et al., 2020). In the studies showing a positive relationship between the peak relative amplitude of bumblebee vibrations and pollen extraction (De

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Luca et al., 2013; Rosi-Denadai et al., 2020), the greater peak relative amplitude vibrations produced by larger *B. impatiens* bumblebees resulted in the removal of more pollen grains per flower visit. Due to their ability to generate higher intensity vibrations for pollen removal, larger *B. morio* bumblebees emitted fewer vibrational pulses per flower visit. This pattern also explains the shorter total vibration time per visit observed in larger *B. morio* individuals. In this study, smaller bumblebees emitted a higher number of vibrational pulses during visits to *C. latistipula* flowers and spent more time to grooming themselves due to repeated grooming periods between pulses, known as non-consecutive grooming behavior. These smaller bumblebees effectively accomplished pollen extraction through multiple alternating vibrational pulse and grooming sequences, encompassing angular displacements and exploration of distinct poricidal anthers of the flower. In alignment with our findings, Vallejo-Marín (2019) found that both large and small bumblebees could achieve similar vibration frequencies during flower interactions.

Body size and behavior complement each other to explain the process of pollen removal and pollination

We expected larger *B. morio* bumblebees to collect more pollen grains from poricidal anthers per flower visit than smaller bumblebees. We also predicted that a larger bumblebee body size and pollen load would be associated with greater numbers of pollen grains transferred to flower stigmas, producing fruits with more viable seeds. However, our results did not support these predictions. Pollen removal and deposition and fruit and seed formation in *C. latistipula* were independent of *B. morio* body size. Surprisingly, a single visit by a *B. morio* bumblebee, regardless of body size, removed nearly all of the pollen grains from the poricidal anthers and guaranteed fruit development. On average, *B. morio* deposited 40 pollen grains on the stigma during a single visit, more than twice the number of ovules in the ovary of *C. latistipula* flower (Delgado, 2019). Based on the quantity of pollen deposition, *B. morio* can be considered an effective pollinator of *C. latistipula* flowers. How do we explain this high pollination performance by both small and large *B. morio* in *C. latistipula* flowers? Smaller *B. morio* bumblebees increased pollen removal and transfer by increasing their movements and time spent in the flower. Smaller bumblebees are expected to be less-effective pollinators due to their small size, even when the separation between floral sexual organs is short (Willmer and Finlayson, 2014; Solís-Montero and Vallejo-Marín, 2017; Mesquita-Neto et al., 2021). However, in cases involving smaller bumblebees, they sequentially touch the stigma due to their longer visit durations, higher angular displacement (movement) within flowers, and a greater number of vibrational pulses per visit. This behavioral compensation overcomes potential limitations in pollen extraction, leading to an increased pollen deposition on the stigma.

We should consider additional factors that could potentially influence pollination, factors that were not controlled in this study and go beyond bumblebee body size and behavior. Among these factors, the most significant is the pre-existing pollen load carried by *B. morio* bumblebees before they visit the *C. latistipula* flowers. Despite the likelihood of bees arriving with pre-existing pollen, controlling *B. morio* pollen loads in field experiments is impractical. Therefore, we sampled the bumblebees on the flowers 1 h after anthesis, assuming that pollen loads carried by the bumblebees were minimal. This assumption was reasonable due to our observations in the recordings, which allowed us to estimate the approximate amount of pollen accumulated in the corbicula of the bumblebees. In addition, we did not observe any effect of the time of bee visit on the number of pollen grains deposited on the stigma or the probability of fruit development. Consequently, we considered these additional factors to be less critical in explaining the observed patterns found in this study.

Pollen removal is affected by both the amplitude and duration of vibrations during a flower visit (De Luca et al., 2013; Rosi-Denadai et al., 2020). We found that the vibration amplitude was correlated with bumblebee visitor body size. However, we did not observe an association between vibration amplitude and pollen removal. About 90% of the pollen grains were removed during a *B. morio* visit, regardless of bumblebee body size. In general, the visiting bumblebees were very effective in collecting pollen from *C. latistipula* anthers due to their ability to make behavioral adjustments according to their body sizes. The greater number of vibrational pulses emitted and the longer total vibration times of smaller bumblebees compensated for the higher amplitude pulses and greater pollen removal of larger bumblebees. In this case, there was a trade-off between vibrational intensity and the total duration of the multiple vibrational pulses during flower visits. More than that, if the pollen removed is on average similar and almost complete in each flower, it is possible that pollen explored and pollen dispersed in the body with the potential to reach the stigma are high, which would help to explain the high pollination efficiency independent of bee body size. The behavior here would be crucial to understanding the outcome similarity of the interaction from the plant point of view.

Why would bumblebees remove all pollen grains from poricidal anthers during a single flower visit? According to optimal foraging theory (OFT; MacArthur and Pianka, 1966), natural selection should favor foraging strategies that maximize benefits while minimizing costs. Consequently, larger bees are more efficient at removing pollen, which could provide greater resources for their larvae within a shorter period. Larger bees collecting pollen more efficiently would help mitigate conflicts with potential competitors and predators (e.g., Gavini et al., 2020) and

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reduce visits to flowers lacking resources (e.g., Klein et al., 2017). Therefore, it is likely that larger bumblebees have lower foraging costs compared to smaller bumblebees. From the plant perspective, if this bumblebee species is the most frequent visitor and has exerted greater selective pressure on floral traits than other visitors and environmental factors, the morpho-behavioral fit could be the result of bumblebee selection and explain the high plant fitness. Indeed, *B. morio* workers over a range of body sizes produce positive outcomes for *C. latistipula* fruit and seed set. However, this result highlights part of the outcome for the female fitness component but not for males. Smaller bumblebee visitors spend two and a half times longer than larger bumblebee visitors in each flower. In other words, larger bees can visit two to three flowers during the period that smaller bees visit just one flower in a *C. latistipula* population. In this scenario, larger bumblebees could be more effective in transferring pollen between flowers, increasing the male plant fitness, in contrast to only considering female fitness (e.g., fruit development). Regardless of the differential impact of the bee size on male fitness, the high reproductive outcome detected here caused by the most frequent pollinator should have its evolutionary causes, if adaptive or not, investigated in the future.

The literature lacks information on the relationship between pollen harvesting by pollinators and pollination efficiency (Minnaar et al., 2019). In our study, *B. morio* extracted a large quantity of pollen from *C. latistipula* flowers in just one visit, regardless of body size. However, it is unknown if the high pollen removal has a high cost for the plant. We can speculate about two potential scenarios: an interaction that results in a low level of pollen loss and another that results in a high level of pollen loss for the plant. Pollen loss would include the transfer of pollen to locations other than the stigma of *C. latistipula*. In the first scenario, the bumblebee removes a high quantity of pollen and receives most of it in safe sites on its body (Macior, 1967; Kimsey, 1984; Thorp, 2000; Koch et al., 2017). This scenario favors the distribution of male gametes (pollen) from the focal flower to the surrounding plant population, especially when pollinators are limited (Minnaar et al., 2019). For the second scenario, the bumblebee removes a large amount of pollen from the anthers but loses a large portion of it while visiting the focal flower and subsequent flower visits or/and actively getting much of this pollen to feed their larvae. In the second scenario, the bumblebee may be very effective in grooming the pollen from its body, storing it in its corbiculae, and later feeding in its colony. However, from the plant perspective, effective grooming behavior increases plant energy costs per visit because the amount of pollen deposited on the stigmas of other conspecific flowers decreases. Although it is currently impossible to determine which of these two scenarios occurs more frequently in pollen fate after *B. morio* visitations, our results provide the foundation for future experimental investigations.

FINAL CONSIDERATIONS

Given that bumblebees show substantial body size variation among worker bees (Alford, 1975), similar to the interspecific variation observed in some systems (as in our study), we emphasize the importance of incorporating intraspecific body size variation in pollination studies. This incorporation becomes particularly relevant in field studies with greater variations in bumblebee body size, as was the case in our study. This investigation becomes even more critical when bee behavior inside the flower is complex (e.g., alternation of buzzing and grooming behavior during a circular movement of bees around poricidal anthers) and very long, reaching more than a minute inside the flower. Our study found a clear link between the body size variation and the complex behavior of bumblebees. What makes the system even more interesting is that although the variation in the bee dimensions is compatible with the distance between the floral sexual organs, allowing the potential arrival of pollen grains on stigmas, not necessarily all bees would be effective in promoting pollination. However, in our case, bumblebees were all equally excellent pollinators regardless of the wide variation in body size and behavior within flowers. This pattern probably happened because smaller bees that should extract less pollen from flowers compensate for it by spending more time buzzing poricidal anthers. If they remove the high amount of pollen that larger bees do, they become covered with pollen and will likely have pollen loads similar to that of larger bees. Smaller bees are also more likely to bump into the stigma by staying longer and moving more inside flowers. Even if the amount of pollen dispersed in their bodies were smaller or in different regions of larger bees, the larger movement and longer handling time would increase the probability of pollen being transferred to the stigma. Therefore, the similar reproductive outcomes generated by bumblebees of different sizes are likely explained by body-size association with behavioral differences. If so, the combined effects of different features of pollinating bees could cause unexpected outcomes in pollination mutualisms. These results highlight potential new directions for investigations, including on intraspecific variation of partner traits in pollination and more broadly on mutualistic interactions in natural systems. Our results emphasize that incorporating the complex effects of partner traits in the outcomes of ecological interactions can be essential to understanding generalized mutualistic relationships and their outcomes.

AUTHOR CONTRIBUTIONS

All authors conceived and conducted the study, designed the methods, and wrote the manuscript. A.N. supervised the study. B.C.B. analyzed the data. B.C.B. and A.N. secured funding. All authors reviewed and edited the manuscript.

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DATA AVAILABILITY STATEMENT

The data sets generated for this study are available on Figshare: https://doi.org/10.6084/m9.figshare.23819373.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. The experimental population of *Chamaecrista latistipula* at the Federal University do ABC (UFABC).

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Appendix S2. Description of the angular displacement of *Bombus morio* bees on flowers of *Chamaecrista latistipula*.

Appendix S3. Oscillogram of one bee visit on flower of *Chamaecrista latistipula* describing the visit time.

Appendix S4. Protocol for counting pollen grains extracted from poricidal anthers of flowers of *Chamaecrista* (Fabaceae).

Appendix S5. Slides and magnified image of *Chamaecrista latistipula* stigma after visiting the bee species *Bombus morio*.

Appendix S6. Relationship among the components of pollination performance and reproductive fitness of *Chamaecrista latistipula*.

Appendix S7. The first three principal components (PCs) calculated over bee behavior descriptors.

Appendix S8. The first two principal components (PCs) calculated over bee behavior descriptors.

Appendix S9. Randomization test over the eigenvalues of the principal components (PCs) of bee behavior descriptors.

Appendix S10. Randomization test over the loadings from the first two principal components (PCs) calculated over bee behavior descriptors.

Appendix S11. The relationship between the intertegular distance of the bee body and the first two principal components (PCs) that described the bee behavior.

Appendix S12. The partial residual plots describing the relationship between the number of pollen grains deposited on the stigma and the first two principal components (PCs).

Appendix S13. The partial residual plots describing the relationship between the fruit formation and the first two principal components (PCs).

Appendix S14. The partial residual plots describing the relationship between the proportion of viable seeds per fruit and the first two principal components (PCs).

Appendix S15. The partial residual plots describing the relationship between the proportion of pollen grains removed by the bee visit from the anthers and the first two principal components (PCs).

Appendix S16. Synopsis of studies on the effect of bee body size on pollination and/or bee behavior on flowers.

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